

ASSESSMENT TOOL-3

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Course Name: Cloud Computing

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

Answer:

Cloud computing is the on-demand delivery of computing resources such as servers, storage, databases, networking, software, and analytics over the internet with pay-as-you-go pricing, instead of owning and maintaining physical infrastructure.

Characteristics of Cloud Computing

1. On-Demand Self-Service

- Users can provision computing resources whenever required without interacting with the service provider.
- **Example:** Launching a virtual machine on AWS within minutes.

2. Broad Network Access

- Cloud services are accessible through the internet using laptops, smartphones, tablets, and desktops.
- **Example:** Accessing Google Drive from any device.

3. Resource Pooling

- The provider pools computing resources to serve multiple customers using a multi tenant model.
- Resources are dynamically allocated based on demand.

4. Rapid Elasticity

- Resources can be increased or decreased automatically according to workload. •

Example: An e-commerce website automatically adding servers during festival sales.

Difference from Traditional On-Premise Computing

Traditional systems require organizations to purchase, install, and maintain hardware in their own premises, whereas cloud computing provides scalable resources over the internet with

minimal upfront investment.

2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

Answer:

Cloud computing evolved through several technological stages.

1. Hardware Evolution (1950s-1970s)

- Computing relied on large mainframe computers shared among multiple users.
- Time-sharing allowed efficient utilization of expensive hardware.

2. Personal Computing Era (1980s)

- Personal computers became affordable.
- Computing shifted from centralized systems to individual machines.

3. Networking and Client-Server Computing (1990s)

- Organizations connected computers using LANs and the Internet.
- Client-server architecture enabled centralized applications and data storage.

4. Virtualization Technology (Late 1990s-2000s)

- Hypervisors enabled multiple virtual machines to run on one physical server.
- Improved hardware utilization and reduced infrastructure costs.

5. Internet Software Evolution (2000s-Present)

- Web applications and broadband internet enabled software delivery over the web.
- Cloud providers like AWS, Microsoft Azure, and Google Cloud introduced scalable computing services.

How These Changes Enabled Modern Cloud Platforms

- Better hardware utilization through virtualization.
- High-speed internet enabled remote access.
- Distributed data centers improved reliability.
- Automation allowed rapid provisioning and scaling of resources.

3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

Answer:

Role of Server Virtualization

Server virtualization divides one physical server into multiple virtual machines (VMs), each operating independently with its own operating system and applications.

Importance in Cloud Computing

- Maximizes hardware utilization.
- Reduces infrastructure costs.
- Enables rapid VM deployment.
- Provides workload isolation.
- Supports scalability and disaster recovery.

Difference between Virtualization and Cloud Computing

Virtualization	Cloud Computing
Technology that creates virtual machines.	Service model delivering computing resources over the internet.
Focuses on efficient hardware utilization.	Focuses on providing on-demand IT services.
Requires hypervisors like VMware or Hyper-V.	Uses virtualization along with automation, networking, billing, and management.
Can exist without cloud computing.	Typically depends on virtualization technologies.
Example: VMware ESXi creating VMs.	Example: Amazon EC2 providing virtual servers as a cloud service.

Conclusion

Virtualization is the **foundation technology**, while cloud computing is the **complete service delivery model** built on top of virtualization.

4. Discuss the role of data centers in delivering cloud services.

Answer:

A **data center** is a secure facility containing servers, storage devices, networking equipment, cooling systems, and power infrastructure that support cloud services.

Roles of Data Centers

1. Compute Resource Hosting

- Houses physical servers that execute cloud applications.

2. Data Storage

- Stores user files, databases, backups, and application data.

3. High Availability

- Uses redundant power supplies, backup generators, and networking to ensure continuous operation.

4. Security

- Provides physical security, surveillance, fire protection, encryption, and access control.

5. Scalability

- Additional servers and storage can be added to handle increasing workloads.

6. Disaster Recovery

- Data is replicated across multiple locations to prevent data loss during failures.

Example

Google Cloud data centers store Gmail data, process Google Search requests, and provide computing services to businesses worldwide.

Conclusion

Data centers form the backbone of cloud computing by providing reliable, secure, scalable, and continuously available computing resources.

5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Answer:

Service Model	Description	Customer Manages	Provider Manages	Real-World

IaaS (Infrastructure as a Service) PaaS (Platform as a Service) SaaS (Software as a Service) XaaS (Anything as a Service)	Provides virtual machines, storage, and networking infrastructure.	Operating system, applications, data	Hardware, networking, virtualization	Example Amazon EC2
	Provides a platform to build, test, and deploy applications.	Application code and data	Infrastructure, OS, runtime, middleware	Google App Engine
	Ready-to-use software delivered through a web browser.	User data and settings	Entire application and infrastructure	Google Workspace
	Any IT service delivered through the cloud, including storage, databases, AI, security, and networking.	Depends on the service	Service infrastructure and maintenance	Microsoft Azure AI Services

Key Differences

- **IaaS:** Highest customer control over infrastructure.
- **PaaS:** Developers focus only on application development.
- **SaaS:** End users simply use the software.
- **XaaS:** Broad category covering all cloud-based services beyond IaaS, PaaS, and SaaS.

Conclusion

These cloud service models provide varying levels of management responsibility, allowing organizations to choose the most suitable solution based on their technical and business needs.